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STUDY PROTOCOL

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Taurine supplementation in children with autism spectrum disorders: a study protocol for an exploratory randomized, double-blind, placebo-controlled trial

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Abstract

Background ASD is the most common neurodevelopment dysfunction and disabling disorder in childhood, with a lack of specific clinical treatment methods. Evidence-based study reveals that nutrients supplement can improve the core symptom of ASD. Our previous study demonstrated the children with ASD were found to have lower taurine levels in serum and urine samples. However, whether or not the taurine supplementation can improve the core symptom of ASD is unclear.

Methods This is an exploratory randomized, double-blind, placebo-controlled trial. Sixty children with ASD will be randomly allocated to receive daily taurine supplementation (taking orally with the recommended dose) or placebo in a 1:1 ratio for 3 months. All children will receive behavioral treatment throughout the study period. All researchers, participants and their caregivers will be blinded until the completion of data analysis, and randomization and allocation will be fully concealed from all mentioned parties. Follow-up will be conducted at the baseline, 1st month, 2nd month, 3rd month, 6th month, 9th month and 12th month. The primary outcomes will focus on changes in scores of ASD-specific questionnaires.

Discussion To our knowledge there are no randomized controlled trials assessing the effects of daily taurine supplementation on the core symptoms of ASD. This trial would help evaluate the efficacy of taurine supplementation on ASD, and initially explore if its potential value on clinical application outweighs the risk.

Trial registrations ClinicalTrials Registry NCT05980520. Registered on July 31, 2023.

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Keywords Autism spectrum disorders, ASD, Taurine, Supplementations

Background

Autism spectrum disorders (ASD) are the most common disabling neuro-developmental disorder. The main clinical features of ASD are social disorders, rigid behaviors, restrictive patterns of behaviors and interests [1]. In recent years, the prevalence of ASD has been increasing year by year. According to the latest epidemiology survey in the United States, the prevalence of ASD is 1/44 [2]. Our previous multi-center epidemiological surveys reveal the prevalence of ASD in children aged 6–12 years in China is about 0.7% [3]. And the lifetime medical and non-medical expenses of patients with ASD ranged from \$1.4 million to \$2.4 million [4]. ASDs have become a major public health problem due to their high prevalence rate, disability rate and heavy disease burden, bringing huge economic and social burden to families and society [5].

Comorbidity with other conditions is often common in children with ASD. Among these comorbidities, gastrointestinal problems are one of the most frequent symptoms. Our previous study showed that 41.4% of children with ASD had gastrointestinal symptoms (GIs) [3]. In animal models, GIs severity was likely to be strongly correlated with ASD symptoms, which are modulated by the intestinal microbiome [6]. Autistic traits such as restricted interests are associated with less-diverse diet; while less-diverse diet, in turn, is associated with lower microbiome alpha-diversity [7]. For another, the gut microbiota also participates in regulating the maturation and function of central nervous system (CNS) immune cells, such as microglia and astrocytes. Microbiota-devoid mice have impaired microglia maturation and function, as well as improper myelin formation [8]. The microbiota also modulates the activation of peripheral immune cells, which play significant roles in neuroinflammation, brain injury, autoimmune responses, and neuro-development [6]. In recent years, the connection between the microbiota-gut-brain axis (involving bidirectional interplay between the brain and the gut including the gut-associated immune system, the enteric neuroendocrine system, the enteric nervous system, and the gut microbiome) and neuropsychiatric disorders has emerged as a major research focus [9, 10]. The complex interplay between the microbiota-gut-brain (MGB) axis and the host may influence oxidative stress in CNS by modulating levels of reactive oxygen species (endogenous and exogenous) and antioxidant systems. Increasing evidence supports the role of the MGB axis in the pathogenesis of ASD [6]. ASD animal models reveal defects in the gastrointestinal barrier, allowing toxins and bacterial products to enter the bloodstream and subsequently affect brain function [6]. Because of an increasing amount of evidence implicating the intestinal microbiome not only in normal development and functioning of the CNS but also as a possible causative or facilitative agent in neuropsychiatric diseases such as ASD, considerable interest exists in nutritional interventions. For instance, children with ASD are more likely to be deficient in nutrients, such as vitamin D, folic acid and unsaturated fatty acids, due to gastrointestinal disease, sometimes because of poor eating behaviors and/or food allergies [11]. On this account, series of clinical studies were performed, focused on the effect of nutrients supplementation including vitamin D, folic acid or unsaturated fatty acids were performed; results showed improvement of the clinical symptoms of ASD [12, 13].

Interestingly, pre-existing non-targeted metabolomics studies using 1 H-NMR have found that urinary taurine levels were lower in children with ASD than in healthy controls [14]. In addition, we use the ultra-high performance liquid chromatography tandem mass spectrometry to detect target serum products of carbon metabolism cycle and confirmed taurine deficiency in children with ASD compared with healthy controls [15]. Taurine is widely distributed in the brain, spinal cord, heart, muscle cells and retina, and is one of the amino acids that are more abundant in almost all tissues and organs [16, 17]. Biosynthesis of taurine occurs in the liver and initiates from methionine via cysteine, and finally excreted via the kidney and the feces [18]. In the CNS, the primary sources of taurine include taurine that directly crosses the blood-brain barrier from the periphery and taurine synthesized within astrocytes (occur in the hippocampus and cerebellum) [19, 20]. Series of researches have showed that taurine play important role in modulation of neurogenesis in the developing brain, neuroinflammation, and neuromodulation [18]. Moreover, numerous studies have shown that taurine has the ability to modulate endoplasmic reticulum stress and Ca^{2+} homeostasis, which have been recognized as one causative factors of ASD and contribution to development of stereotypical behaviors [21]. In children with ASD, gastrointestinal issues may lead to reduction of the taurine endogenous synthesis and dietary intake. Additionally, animal studies have demonstrated that exposure to environmental risk factors can impair astrocyte function, further decreasing taurine synthesis in the CNS [22]. Both could lead to decreased sources of taurine in the CNS. However, most peripheral taurine exists in free form outside the CNS, with only a small fraction entering the brain [18]. As a crucial component of the bile acid metabolism, peripheral taurine has been linked to abnormal bile acid metabolism observed in ASD patients [23]. Bile acid metabolism interacts with gut microbiota, which is a crucial component of the MGB axis. Taken together, taurine is likely associated with ASD via MGB axis.

Recently, taurine has been studied on several animal-models, including depression and anxiety, neurodegenerative diseases, epilepsy, and stroke, and showed some treatment activity [18]. Regarding to ASD, despite there is no clinically study assessing effect of taurine supplementation in ASD individual, animal model study has showed exciting results. Taurine supplementation can increased the proliferation of hippocampal cells, and reduced phosphorylation of mammalian target of rapamycin (mTOR) in hippocampal tissue of the BTBR T + tf/J (Black and Tan Brachyury, BTBR) mice, and further improved the ASD-like behavior [24]. In addition, in-silico molecular docking study suggested taurine might be a potential adjunct therapy in ASD [25]. Importantly, clinical trial indicated that the majority of participants with succinic semialdehyde dehydrogenase (SSADH) deficiency [18] showed changes in the social, conceptual, and practical adaptive behavior scores after taurine therapy [26]. However, whether taurine supplementation can improve the core symptoms of ASD children and its mechanisms need to be further investigated. The objectives of this study are: (1) to investigate the efficacy of taurine supplementation in improving core symptoms in children with ASD; (2) to explore if its potential value on clinical application outweighs the risk, by observing if any adverse events occur to the taurine intervention and the adherence.

Aim of study

This exploratory trial aims to evaluate whether children with taurine supplementation is superior to those who receive placebo on better rehabilitation of ASD core symptoms and comorbidities.

Study setting

Study site is the pediatric outpatient clinic of Guizhou Hospital of Shanghai Children's Medical Center, a children's general hospital in Guizhou Province, China. This is a public children's hospital that fully equipped and routinely provide treatment for children with ASD.

Trial design

This is an exploratory randomized, double-blind, placebo-controlled trial. A total of 60 children (aged 2.5–18 years) will be recruited and followed-up for 12 months. On the first day after enrollment, demographic information is collected, relevant laboratory tests and assessment scales will be performed. Interventions including dietary supplementation and behavioral therapy as usual will be provided since the second day of enrollment. Subjects will be randomized to take either daily taurine supplementation or placebo for 3 months, while behavioral therapy will be continued as usual until the end of study

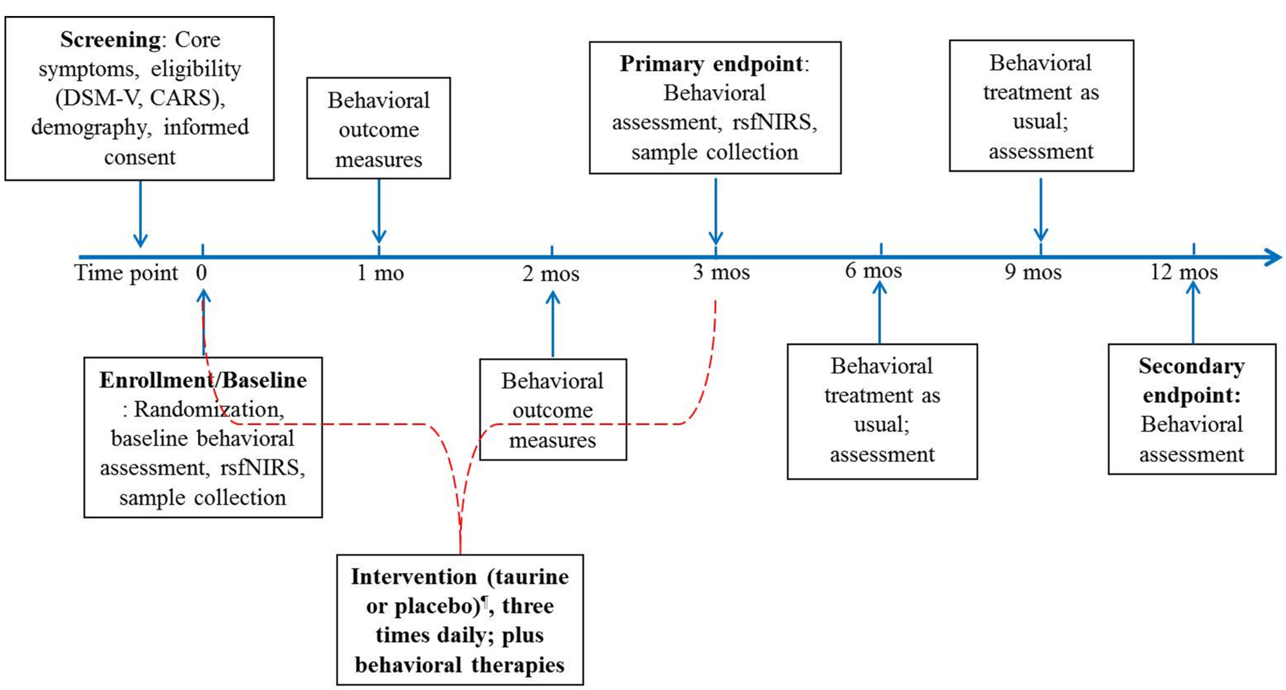

Fig. 1 Study procedure. DSM, Diagnostic and Statistical Manual of Mental Disorders rsfNIRS; CARS, Childhood Autism Rating Scale; rsfNIRS, resting-state functional near-infrared spectroscopy; mo, month. Treatment begin from the second day of enrollment

period (follow-up 9-month after 3-month intervention). This trial has been approved by the research ethics board of Guizhou Provincial People's Hospital [No.2023-046]. Study procedure sees Fig. 1.

Recruitment

Screening and recruitment will take place through offline in outpatient clinics in Guizhou Hospital of Shanghai Children's Medical Center. All individuals will fit the study's inclusion and exclusion criteria. The purpose, significance and specific content of the study will be explained in detail to the subjects and their legal guardians by researchers. Written consent will be obtained from the subjects' legal guardians before randomization. All information will be confidential and only be accessible to researchers who involved in recruitment process.

Eligibility criteria

Inclusion criteria

1. Age: 2.5 ~ 18 years old at enrollment, male or female;
2. Diagnosis made by a professional doctor based on the diagnostic criteria for ASD of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5). The CARS-2 is used to support a clinical diagnosis of ASD.
3. The legal guardian fully understands and provides the written consent.

Exclusion criteria

1. Children diagnosed with Rett syndrome, angel syndrome, cerebral palsy, epilepsy, other genetic metabolic diseases;
2. Children who have had an acute or chronic infectious disease in the past 3 months;
3. Children with abnormal liver and kidney function or chronic pulmonary heart disease;
4. Children who have taken any supplements containing taurine or medications in the past 3 months;
5. Parents or patients refuse to participate in the study;
6. The investigator considers that the subject is unable to comply with the study requirements;
7. Refuse to consume the supplementation lasting to more than one week.

Interventions

Rationale for the choice of supplements

Corn starch and white sugar (proportion of 5:1) will be the main ingredient of placebo, which is similar to another taurine supplementary trial [27]. The placebo has no detrimental or disturbing effect on core syndromes of ASD, and in good taste that will increase acceptability in children with ASD. The taurine supplements also are mixed with corn starch and white sugar, with identical appearance, taste and smell as that in the control group, except that experimental supplementation contains taurine. Taurine supplements and placebos are prepared by an independent team prior to the start of recruitment.

Intervention description

1. Experimental group: Taurine Supplement + Behavioral therapy.
   The experimental supplementation will be taurine (Hubei Yuanda Life Science and Technology Co., LTD) mixed with corn starch and white sugar (Taurine 0.4 g + Corn starch 0.1 g + White sugar 0.1 g) in 1 bag, taken orally. Subjects will be instructed to have supplementations with specified dosages and frequency: 1 bag each time for 1–2 years old, 3 times a day;1.5 bags each time for 3–5 years old, 3 times a day;2 bags each time for 6–8 years old, 3 times a day;2.5–3.5 bags each time for 9–13 years old; 3 to 4 bags each time for children over 14 years old, 3 times a day. The dosage for different age groups is approximately 100–200 mg/kg.d. According to the previous studies, such dosage range of taurine can reach the effective plasma concentration of taurine, with mild treatment-related adverse events [26, 28–30]. The use of taurine is strictly in accordance with the specifications of Pharmacopoeia of the People's Republic of China [31]. The supplementations will be taken for 3 months since enrollment
   Behavioral therapy is structured courses with the Applied Behavior Analysis (ABA), which emphasizes play, social interaction, and communicative initiation on the part of the child, and natural consequences as opposed to rewards such as food [5]. The treatment is generally undertaken by a therapist working one-to-one with a child, using principles of learning to teach a child developmental skills such as language, imitation, or cognitive tasks such as matching or sorting. The treatment is usually intended to be given intensively in periods of 3–4 h daily, sum up with 15–20 h weekly [5, 32].

2. Control group: Placebo Supplement + Behavioral therapy.
   The placebo will be mixture of corn starch (0.5 g) and white sugar (0.1 g), with identical dosages and frequency as experimental group within the same age range. Behavioral therapy will be performed as same as the experimental group.

3. Availability of supplements: These supplements will be given to the parents in the clinic during the child's

monthly visit to the hospital. Parents will be given one month's dosage and be told about the dosages of the supplements. The mealtimes of consumption is not recommended in this trial.

4. Throughout the entire study period, other supplementations containing taurine and probiotics will be strictly prohibited. Participants' dietary habits will be asked to be maintained as usual and recorded in diaries. Medication use and other disease status during the study will be recorded in detail in the case report form (CRF).

Adherence

The adherence monitoring of supplementations will be undertaken at clinical visits (once a month). Participants will be assessed the compliance with supplementation consumption monthly according to the following rationale: (1) high compliance: follow the instruction and take the supplementation with specified dosages and frequency; (2) moderate compliance: fail to take the supplementation with specified dosages and frequency ≤ 2 times per week; (3) poor compliance: fail to take the supplementation with specified dosages and frequency > 2 times per week. For those with moderate or poor compliance, the guardians will be encouraged to stick to consuming supplementation. Other strategies to improve adherence include: (1) subjects' parents or caregivers will be asked to write diaries to record the compliance of interventions; (2) regular reminding will be provided by research staffs; (3) offering group consultations for the families during structured training courses; (4) keeping regular follow-up by phone and the WeChat application.

Safety

Subjects will be allowed to withdraw from the trial at any time with no reason. The incidence of adverse events regarding taurine supplementation is rarely reported previously, but events like itching, nausea, headache, vertigo, gait disturbance, hypothermia, hyperkalemia, gastric ulcers may occur during taurine use, which are usually spontaneously relieved after discontinuation. Main adverse events will be recorded to evaluate the safety of interventions at every follow-up visit, or reported by subjects or their guardians throughout the study period, including the occurrence time, symptoms, signs, duration, laboratory examination indicators, treatment methods, results, follow-up time, etc. Severity will also be recorded according to the following criteria: (1) normal: no discomfort; (2) mild discomfort: does not affect daily life and work; (3) moderate discomfort: affect daily life and work to some extent; (4) severe discomfort: significantly affecting life and good condition (bed rest required); (5) extremely severe: life-threatening. Adverse events will be collected systematically in the same manner for each participant and recorded in the CRF. Unexpected harms will also be collected in the CRF. All the events will be reported in trial publications and the original CRF will also be uploaded to ClinicalTrials.gov if necessary. To monitor the health condition of subjects, routine blood examination and urinalysis will be performed at baseline and 3rd month.

The correlation between adverse events and the interventions will be evaluated and recorded in the CRF. Investigators will address the signs and symptoms of the subjects in a timely manner, and consider discontinuation or withdraw subjects with severe adverse events from the study if necessary. If one child is unable to consume the supplementation lasting to more than two weeks in the first month due to adverse reactions or other factors (such as dislike consuming the supplementation), he/she will be withdrawn from further procedure of the study. Secondly, if one child stops consuming the supplementation lasting to more than two weeks in the second or third month, he/she will be still included in the subsequent analysis. All the situations mentioned above will be carefully documented in the CRF. Furthermore, comprehensive accident insurance was purchased for each child.

Study outcomes

Outcomes measures will include questionnaires, anthropometric measurement, brain function and biochemical tests. Questionnaires, anthropometric measurement, brain function will be performed at every follow-up visit (i.e. at baseline, 1 st, 2rd, 3rd, 6th, 9th, 12th month). Biochemical tests will be performed at baseline, 3rd month and 12th month. A brief summary of outcome measures is showed in Table 1. All outcomes will be evaluated during the follow-up period according to the pre-determined timeline (Fig. 1; Table 2).

Primary outcomes

Considering that the treatment endpoint is 3 months after treatment, we specified the Autism Treatment Evaluation Checklist (ATEC) score as the primary outcomes measure, which will be assessed at the baseline, 1 st month, 2nd month, and 3rd month, respectively. ATEC is a questionnaire that was developed by the Autism Research Institute to evaluate the treatment efficacy in autistic individuals [33]. The ATEC is suitable for ASD children 2–18 years of age. It consists of four subscales: speech/language/communication (14 items, total scores range from 0 to 28), social interaction (20 items, total scores range from 0 to 40), sensory/perception (18 items, total scores range from 0 to 36), and health/physical/behavior (25 items, total scores range from 0 to 75) [34]. The four subscale scores can be used to calculate a total score (range from 0 to 179 scores). The higher the scores, the more severity of the ASD symptoms. Based on the

Table 1 Summary of the study outcome measures and methods

Contents	Methods	Age limit(years)	Assessor
Rehabilitation of ASD	ATCE	2–12	By caregivers
Central nervous system function	Gesell	≤6	By clinicians
Central nervous system function	WISC	>6	By clinicians
Assess the severity of autism	CARS	>2	By clinicians
Severity of ASD	ASRS	2–18	By caregivers
Social skills	SRS	4–18	By caregivers
Severity of eating behavior problems	IMFeD	All ages	By caregivers
Severity of gastrointestinal symptoms	6-GSI	All ages	By caregivers
Sleep habits	CSHQ	3–12	By caregivers
Monitoring brain function	fNIRS	All ages	By clinicians
Nutritional status	BMI	All ages	By nurses
Intestinal flora analysis	Stool	All ages	By laboratorians
Markers of oxidative stress, bile acid metabolites, serum taurine content	Blood	All ages	By laboratorians
Urine taurine content	Urine	All ages	By laboratorians

Abbreviations: *ASD* autism spectrum disorders, *ATEC* autism treatment evaluation Checklist, *WISC* Wechsler intelligence scale, *CARS* childhood autism rating Scale, *ASRS* autism spectrum rating Scales, *SRS* social responsiveness Scale, *IMFeD* identification and management of feeding Difficulties, *6-GSI* six Gastrointestinal severity Index, *CSHQ* children's sleep habits Questionnaire, *fNIRS* functional near-infrared spectroscopy, *BMI* body mass index

scores, Severity can be classified with mild (initial ATEC score 20–49), moderate (initial ATEC score 50–79), severe (initial ATEC score >80). This measurement was parent/caregivers-reported and completed through an online service. The main outcomes will be the difference between the two groups in the ATEC scores after 3 months of treatment.

Secondary outcomes

The secondary outcomes include the difference between the two groups in the Childhood Autism Rating Scale (CARS), the Social Responsiveness Scale (SRS), the Autism Spectrum Rating Scales (ASRS), the Identification and Management of Feeding Difficulties (IMFeD), the Six Gastrointestinal Severity Index (6-GSI), and the Children's Sleep Habits Questionnaire (CSHQ) scores. The other outcomes including the Gesell score, The Wechsler intelligence scale (WISC) score. These scales will be assessed at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively. In addition, we will assess the ATEC scores at the 6th month, 9th month, and 12th month. In addition, we will investigate the resting-state brain function in the frontal lobe by functional near-infrared spectroscopy, and perform fecal meta-genomic sequencing and blood untargeted metabolomic analysis to identify the potential mechanism.

1. The Childhood Autism Rating Scale (CARS)
   The CARS is designed to measure ASD severity for children over 2 years old, and need to be evaluated by professional clinicians. The CARS consists of 15 items with a total score of 60 points; the number of items that assigned a score ≥ 3 was also used to measure the severity of the symptom. The total score is 60 points, and a score of 15–30,31–36,37–60 are considered as mild autism, moderate autism, severe autism respectively [35]. CARS will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.
2. The Social Responsiveness Scale (SRS)
   SRS are used to assess social ability for 4–18 years old children. It consists of 65 items, which includes five subscales: social awareness, social cognition, social communication and social motivation and autistic mannerisms [36]. SRS will be self-assessed by parents or other caregivers. The score is calculated and converted into a total score (range from 0 to 195 scores), which is divided into normal (total scores range from 0 to 60), mild (total scores range from 60 to 75), moderate (total scores range from 75 to 90) and severe (total scores range beyond 90). The higher the scores, the more severity of the social symptoms. SRS will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.
3. The Autism Spectrum Rating Scales (ASRS)
   ASRS will be self-assessed by parents or other caregivers for 2–5 years old children (70 items) and 6–18 years old children (71 items) [37, 38], covering defects in the nature of social interaction, abnormalities in language and communication, stereotypes, limitations, repetitive interests and behaviors. The ASRS questionnaire uses a 5-level score: 0 = never, meaning that it has never happened; 1 point = very few, refers to 1 or 2 times in a month; 2 points = sometimes, refers to 1 or 2 times in a week; 3 points = frequent, meaning 3 or 4 times in a week; 4 points = always, which means every day. ASRS will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.
4. The Identification and Management of Feeding Difficulties (IMFeD)
   IMFeD will be self-assessed by parents or caregivers, which composed of 17 items and 6 dimensions: poor appetite, particular food preferences, poor eating

Table 2 Schedule of enrolment, intervention, and assessment

	Enrolment	**Allocation**	**Post allocation**						
Timepoint	**−2 to 0 weeks**	**0 week**	**Baseline**	**1month**	**2month**	**3month**	**6month**	**9month**	**12month**
Enrolment：									
Informed consent	×								
Demography	×								
Eligibility	×								
Allocation		×							
Interventions:									
Taurine			×	×	×	×			
Placebo			×	×	×	×			
Behavioral rehabilitation therapy			×	×	×	×	×	×	×
Adherence and safety：									
Lifestyle habit (Diary)			×	×	×	×	×	×	×
Compliance			×	×	×	×	×	×	×
Adverse events			×	×	×	×	×	×	×
Routine blood examination			×			×			×
Urinalysis			×			×			×
Outcome collection：									
Gesell/WISC			×	×	×	×	×	×	×
ATEC			×	×	×	×	×	×	×
CARS			×	×	×	×	×	×	×
SRS			×	×	×	×	×	×	×
6-GSI			×	×	×	×	×	×	×
IMFeD			×	×	×	×	×	×	×
CSHQ			×	×	×	×	×	×	×
Brain function			×	×	×	×	×	×	×
BMI			×	×	×	×	×	×	×
Intestinal flora analysis			×			×			×
Blood biochemistry			×			×			×
Oxidative stress markers			×			×			×
Serum taurine content			×			×			×
Urine taurine detection			×			×			×

Abbreviations: *ATEC*, Autism Treatment Evaluation Checklist, *WISC* Wechsler intelligence scale, *CARS* Childhood Autism Rating Scale, *ASRS* Autism Spectrum Rating Scales, *SRS* Social Responsiveness Scale, *IMFeD* Identification and Management of Feeding Difficulties, *6-GSI* Six Gastrointestinal Severity Index, *CSHQ* Children's Sleep Habits Questionnaire, *fNIRS* Functional near-infrared spectroscopy, *BMI* Body mass index

habits, excessive parental concern, fear of eating, and underlying disease status [39]. The severity of children's eating behavior problems is divided into four levels: "Always" refers to an average of 5 or more days per week when the eating behavior or feeling; "Often" means having this behavior or feeling on average 3 to 4 days per week; "Sometimes" means having this behavior or feeling on average 1 to 2 days per week; "Never" means that the action or feeling does not occur. IMFeD will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.

5. The Six Gastrointestinal Severity Index (6-GSI)
6-GSI is used to assess the severity of gastrointestinal symptoms, including constipation, diarrhea, whether the stool is formed, abnormal stool odor, abdominal distension and abdominal pain [40]. The severity of each symptom is assessed according to the child's situation in the past 2 to 3 months (0 to 2 scores), and those who have at least 1 symptom scored 2 points are recorded as "having gastrointestinal symptoms". The higher the score, the more severe the syndromes. 6-GSI will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.
6. The Children's Sleep Habits Questionnaire (CSHQ)
CSHQ assesses eight specific sleep problems, including bad sleeping habits, sleep anxiety, irregular sleep duration, sleep disordered breathing, abnormal sleep, daytime sleepiness, night waking, and prolonged sleep latency. Parents or caregivers will need to recall their children's sleep conditions in the past four weeks, and rated the children's sleep conditions as "often" (5–7 times per week),

"sometimes" (2–4 times per week), and "rarely or none" (0–1 times per week) [41]. CSHQ will be self-assessed by parents or caregivers, at the baseline, 1 st month, 2nd month, 3rd month, 6th month, 9th month, and 12th month, respectively.

7. The Gesell score
The Gesell developmental scale mainly assesses the function of the central nervous system and has diagnostic value for autism and developmental delay, which will be evaluated by professional clinicians. It has five specific domains: gross movement, fine movement, language, adaptation and individual-social behavior. According to the relationship between the scores obtained in five behavioral fields and actual age, the development quotient (DQ) of each field will be calculated as following: borderline state: 76 ≤ DQ ≤ 85; mild disorder: 55 ≤ DQ ≤ 75; moderate disorder: 40 ≤ DQ ≤ 54; severe disorder: 25 ≤ DQ ≤ 39; Very severe disorder: DQ < 25. The higher the DQ, the better the development [42].
8. The Wechsler intelligence scale (WISC)
WISC consists of 6 verbal subtests, namely general knowledge, similarity, arithmetic, vocabulary, comprehension, and number memorization [43]. WISC will be evaluated by professional clinicians, including 6 operation sub-tests, namely picture complement, picture arrangement, building block pattern, object matching, decoding, maze. A score of ≥ 130 are very excellent, 120 to 129 are excellent, 110 to 119 are upper middle, 90 to 109 are medium, 80 to 89 are lower middle, 70 to 79 are borderline; a score of < 70 are low intelligence, of which 55 to 69 are mild mental retardation, 40 to 54 are moderate mental retardation, and 25 to 39 are severe mental retardation.

Additional, to evaluate the brain function, we will perform functional near-infrared spectroscopy at baseline and 3rd month, respectively. Functional near-infrared spectroscopy (fNIRS) is a new non-invasive neuroimaging technique for real-time monitoring of brain function. Based on the difference of absorption rate of oxygenated hemoglobin (HbO) and deoxyhemoglobin (DHb) in brain tissue for near-infrared light of different wavelengths, the changes of hemoglobin concentration in cortical metabolism will be calculated to reflect changes of brain function. In the study, we will investigate the resting-state brain function of the frontal lobe by fNIRS.

Moreover, we will perform fecal meta-genomic sequencing and blood untargeted metabolomic analysis to identify the potential mechanism (methodology see Supplementary). The microbiota analyses will be performed by Yunsheng Biotechnology Co (Shanghai, a company engaged in technology development and technical services in the field of biotechnology). Blood samples will be collected for oxidative stress marker detection, bile acid metabolites detection and taurine content determination. Urine sample will also be collected for urine taurine detection. Specimen will be collected at the baseline and 3rd month. Finally, the proportion of adverse events in both groups will also be collected in CRF at every follow-up visit.

Randomization and blinding

Block randomization with 1:1 ratio will be used to assign either taurine supplementation or placebo to enrolled subjects. The randomization sequence is generated by the independent statistic team at the Clinical Trials Unit of Children's Hospital of Fudan University, using the Stata 15.1 software (random seed number: 20230809). To ensure allocation concealment, each allocation sequence was placed in small, opaque, and sealed envelopes numbered and marked in order from one to four and were enclosed in a larger, opaque, and sealed envelope marked with the block number. The clinicians involved in this trial assess eligibility, enroll the subject after obtaining informed consent, and then contact the clinic coordinator who is independent of other research processes to obtain the allocation protocol for that subject. Block envelopes and the four enclosed small envelopes will be disclosed in order by the clinic coordinator.

This trial will be double-blinded where participants and their parents or caregivers, researchers involved in enrollment, assessment, data management and data analysis will not be informed of the allocation during the whole study period. Identity code, rather than highlighting the group allocation on the package (i.e. A or B intervention), is applied for dispensing the supplementation to subjects so as to minimize subjects' discomfort. In an emergency situation where unblinding will be required, the principal investigator will contact the clinic coordinator who is responsible for allocation to reveal the treatment assignment for a given participant by matching both name and the identity code.

Sample size

Considering that there is no previous literature data available for reference for the calculation of sample size, this exploratory trial is expected to recruit 60 subjects (30 for each group) that meet the regular outpatient workload in the study site, which will be expected to reach the size within a year.

Data collection and management

Collection of outcomes, such as clinician-assessed questionnaires, and anthropometric measurement will be originally recorded in written CRF and then entered into Microsoft ACCESS database. Double data entry and check will be completed on a wireless-enabled laptop by two trained operators who will not be involved in other procedures during the trial. Parent-assessed questionnaires and laboratory outcomes will be directly exported from electronic form via Microsoft EXCEL with pre-specified identity codes. The database will be coded and stored in the safe location that is only accessible to personnel involved in data collection. In ACCESS database, there are modules that allow for the identification of missing data, as well as quality assurance mechanisms. The data will be checked for data quality routinely and locked up when the last patient completes follow-up to achieve outcomes (i.e. 12months follow-up).

This single-center trial will be primarily conducted by Department of Pediatrics in Guizhou Hospital of Shanghai Children's Medical Center, with statistical support from Clinical Trial Unit (CTU) of Children Hospital of Fudan University. An internal data monitoring committee (DMC) will be established and will consist of professional pediatricians from Department of Pediatrics and statistical experts from the CTU team, and a senior statistician from different institute for monitoring data completeness, safety information, adverse events, and so forth. Group conferences will be taken place bimonthly for further issues. No auditing will be performed through a professional organization.

Once the subjects are enrolled, retention efforts will be made to minimize loss-to follow-up during the study, such as: 1) all the measurements will be free charged for all participants during the study period; 2) to train and inform subjects and their families about the potential hazards associated with the interventions, understanding how this trial may help their children, and increase the confidence of subjects and their families to complete the trial; 3) To maintain a positive relationship with the subjects, using various means of communication such as the WeChat, SMS, and telephone. Regular contact should be kept with a friendly attitude to establish a harmonious and trustworthy relationship.

Statistical methods

Group comparison of baseline characteristics will be performed as mean (± standard deviation) for continuous variables with normal or approximately normal distribution; or median with interquartile range (IQR) for variables with skewed distribution. Categorical variables will be presented as proportions.

Primary analysis will use all randomized participants regardless of protocol adherence. Repeated measures analysis, such as generalized linear mixed model (GLMM) will be used to determine changes from baseline over time for both primary outcome and secondary continuous outcomes.

Sensitivity analyses will be conducted using per-protocol population, with further adjustment on age, sex and baseline taurine concentration. Subgroup analysis will be considered according to severity of ASD, sex or different age group etc. Multiple imputation for handling missing data will be considered if complete case analyses are not supported. Other detailed strategies will be described in statistical analysis plan and uploaded on ClinicalTrials.gov before the completion of this trial. Interim analysis will not be considered in this trial.

Statistical analyses will be performed using the R4.1.2 (Vienna, Austria) or the Stata 16.1 software (Stata Corp, Texas, USA), and a two-sided alpha level of 0.05 as a statistically significant difference.

Ethics and dissemination

The trial protocol (V.2.0, 20 June 2023) is following the principles of the Declaration of Helsinki, and has been approved by the research ethics board of Guizhou Provincial People's Hospital [No.2023-046]. It has been registered on 31 July 2023 at ClinicalTrials.gov (NCT05980520). Any modifications to the protocol which may impact on the conduct of the study, potential benefit of the patient or may affect patient safety, including changes of study objectives, study design, patient population, sample sizes, study procedures, or significant administrative aspects will require a formal amendment to the protocol. Such amendment will be approved by the research ethics board of Guizhou Provincial People's Hospital prior to implementation and notified to the health authorities in accordance with local regulations. All project team members will be given access to the cleaned data sets, which will be password protected. To ensure confidentiality, data dispersed to project team members will be blinded of any identifying participant information. Trial results will be published in peer-reviewed journals and will be disseminated to the media and the general public. No additional participant data or biological specimens will be collected that requires further consent.

Reporting of research protocol

This protocol is reported according to the SPIRIT 2013 reporting guideline [44]. The checklist is filled out in the supplementary information of additional file 1. All items in accordance with the World Health Organization Trial Registration Dataset have been covered in ClinicalTrials.gov (NCT05980520, Registered on July 31, 2023), with a summary of the content shown in supplementary information of additional file 2.

Discussion

The treatment of ASD does not have specific drugs to improve the core symptoms. Although effective behavioral interventions can significantly improve the core symptoms of children with ASD, among those a number of them still have functional impairments. Therefore, the research and development of new treatment methods for children with ASD has become a hot but challenging issue. Taurine is closely associated with the development of neurodevelopmental disorders; taurine supplementation has a positive regulatory effect on intestinal flora and may improve symptoms of neurodevelopmental disorders. Whether taurine supplementation can improve clinical symptoms of ASD by restoring intestinal homeostasis and reducing oxidative stress in children with ASD needs further study.

With the support of the National Health Commission, our project team established a national multi-center cohort study of ASD for children and constructed a biobank of pediatric ASD. Our case-control study using 1 H-NMR metabolomics to detect urine samples from children with ASD showed lower taurine levels in the ASD group compared to healthy controls [14]. Besides, in previously pilot animal intervention experiment, we have found that taurine supplementation on the basis of VPA autism model in rats during pregnancy could improve ASD-like behavior in offspring (unpublished data). Despite the mechanism of the improvement of taurine supplementation on ASD is unclear, these above-mentioned interesting findings drive us to conduct this exploratory trial for initially investigate the potential value of taurine supplementation on ASD treatment. Unfortunately, diet analyses such as dietary intakes will not be performed in this trial, due to difficulty of collecting detailed dietary information in children with autism. However, we collected the children's dietary profile (IMFeD) in the form of a scale for instead. We expect that this trial will provide important insights into the causality of reported associations. To our knowledge, there are no other randomized controlled trials investigating the effect of taurine on ASD symptoms so far.

It would be meaningful if this study could establish that taurine supplementation can improve symptoms in children with autism, and that the improvement in one area was found to be more prominent in the study. At the same time, the incidence of possible adverse reactions of long-term oral taurine was observed, and the curative effect was evaluated comprehensively. These data will be helpful for subsequent relevant studies.

This exploratory randomized controlled trial (RCT) protocol has several limitations that should be considered. Firstly, the trial is conducted at a single center, which may limit the generalizability (external validity) of the findings to the broader population of children with ASD, who often exhibit significant heterogeneity in symptoms, comorbidities, and genetic backgrounds. Secondly, although participants are instructed to maintain their usual dietary habits and record them in diaries, detailed quantitative dietary intake assessment (e.g., weighing food records or multiple 24-hour recalls) is not performed. This makes it difficult to fully control for potential confounding effects of dietary patterns and nutrient intake (e.g., pre-existing taurine intake from food sources) on the outcomes. Additionally, the exclusion criteria (e.g., children with epilepsy, other genetic metabolic diseases, and abnormal liver/kidney function) are necessary for safety but may result in a study sample that is not fully representative of the broader ASD population, which often has high rates of these comorbidities. This affects the applicability of the results to more medically complex individuals with ASD. Lastly, the taurine supplementation period is 3 months. While there is a follow-up period up to 12 months, the long-term persistence of any potential benefits of taurine supplementation beyond the intervention period remains uncertain and would require longer-term studies.

Abbreviations

CTU	Clinical Trial Unit
ATEC	Autism Treatment Evaluation Checklist
ASD	Autism spectrum disorders
CARS	Childhood Autism Rating Scale
ASRS	Autism Spectrum Rating Scales
SRS	Social Responsiveness Scale
IMFeD	Identification and Management of Feeding Difficulties
6-GSI	Six Gastrointestinal Severity Index
CSHQ	Children's Sleep Habits Questionnaire
DSM-5	Diagnostic and Statistical Manual of Mental Disorders
DQ	Development quotient
WISC	Wechsler intelligence scale
fNIRS	Functional near-infrared spectroscopy
HbO	Oxygenated hemoglobin
DHb	Deoxyhemoglobin
BMI	Body mass index
DMC	Data monitoring committee
ITT	Intention-to-treat analysis
GLMM	Generalized linear mixed model

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12887-025-06216-0.

Additional file 1. Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) 2013 checklist: recommended items to address in a clinical trial protocol and related documents

Additional file 2. Items from the World Health Organization Trial Registration Dataset

Provenance and peer review

Not commissioned; externally peer review.

Biological specimens

Samples of blood (3-5 ml*3 tubes per visit), urine (5-10 ml*1tube per visit) and stool (10 g*1 per visit) will be collected at baseline, 3rd month and 12th month, which are included in the consent form and will be informed by guardians. Detailed plan for collection, laboratory evaluation, and storage of biological specimens will be recorded in the research protocol and uploaded in ClinicalTrials.gov and/or peer-reviewed journal if necessary.

Authors' contributions
HZ, WY, and YW conceptualized the project and developed the protocol. YW and WH provided additional expertise in the statistical analysis. YC, WH, and QD drafted the manuscript. All authors discussed, read, revised the manuscript and all approved the publication of this. The authors are all project team members and no professional writer is hired in this study.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Has been obtained from Guizhou Provincial People's Hospital [protocol number (2023)042, V2.0; dated June 2023]. Recruitment of this trial started on 17 August 2023 and the intervention started on the second day of enrollment (i.e. 18 August 2023). The approximate date when recruitment will be completed is April 2024.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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